

Empirical Analysis of Deterministic Factorization Methods
Systematic Evaluation of Classical and Alternative
Approaches

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.1 Empirical Tests of Period-Finding Methods

Summary

This document describes a test series on integer factorisation via period finding. Classical methods (trial division, Pollard ρ) are tested against resonance-based search methods with different grid widths. The result places the grid parameters correctly: they are numerical sampling quantities, not physical resonance parameters.

Separate parameters

The grid width of the search methods is denoted σ here (values $1/42$, $1/100$, $1/1000$, $1/100000$). It is *not* the FFGFT geometry parameter $\xi = 4/30000$. Earlier versions used the same symbol for both; the separation is necessary because the quantities are unrelated.

.2 Test Setup

Methods tested

Designation	Principle	Parameter
trial_division	Trial division up to $\sqrt{N}$	—
pollard_rho	Cycle detection (Floyd)	—
period_classic	Sequential period search	$\sigma = 1/100000$
period_universal	Sequential period search	$\sigma = 1/100$
period_medium	Sequential period search	$\sigma = 1/1000$
period_special	Sequential period search	$\sigma = 1/42$

The grid width σ determines the step size in which the search space is traversed. It bears no relation to physical eigenfrequencies.

Test categories

1. **Small semiprimes** ($N < 1000$): complete factorisation expected, runtime comparison.
2. **Medium semiprimes** ($10^3 < N < 10^6$): scaling behaviour.
3. **Twin-prime products** ($N = p(p + 2)$): behaviour with structurally adjacent factors.
4. **Large semiprimes** ($N > 10^6$): expected failure of period search.

.3 Results

Small semiprimes

All methods factorise completely. Trial division is fastest at these sizes; period search carries no advantage here, since the effort of searching for r exceeds that of direct division.

Scaling behaviour

[K] With growing N the success rate of period search declines. The reason is the number of candidates to be distinguished: the period r divides $\lambda(N) = \text{lcm}(p - 1, q - 1)$ and can take values of order N . A sequential search therefore requires $O(N)$ steps – considerably worse than trial division with $O(\sqrt{N})$.

Influence of the grid width

[K] Different σ values yield different success rates. These differences are *sampling artefacts*: a finer grid finds more candidates but costs more steps. There is no “optimal” σ in a physical sense – only a trade-off between resolution and effort.

Why σ generates no resonances

The bicoherence test on the Riemann zeros (Doc. 316, August 2026) shows: genuine phase coupling occurs exclusively at prime powers ($b^2 = 0.76\text{--}0.85$), not at composite numbers ($b^2 = 0.018\text{--}0.025$). All σ values used have composite denominators ($42 = 2 \cdot 3 \cdot 7$, $100 = 2^2 \cdot 5^2$, $1000 = 2^3 \cdot 5^3$). In the spectrum they form *mixed points* without a coupling line of their own. Their effect is purely numerical.

Twin-prime products

For $N = p(p + 2)$ the factors lie close together. Trial division benefits (a find near $\sqrt{N}$), period search does not – the period r depends on $\text{lcm}(p - 1, p + 1)$ and shows no particular pattern. The “twin-prime optimisation” via a special grid width described in earlier versions is not reproducible.

.4 Placement

Comparison of methods

Method	Principle	Closure condition	Complexity
Trial division	Direct division	Discrete arithmetic	$O(\sqrt{N})$
Pollard ρ	Cycle detection	Birthday paradox	$O(N^{1/4})$
Period search	Spectral search	Rationalised grid	$O(N)$, Weyl limit
Shor's QFT	Quantum interference	Discrete phase	$O((\log N)^3)$ on QC

[K] Classical period search is inferior to established methods. It has no complexity advantage; the previously claimed agreement with Shor's $O((\log N)^3)$ applies only to the quantum case and is not established for a classical method.

The structural limit

[K] Beyond practical inferiority there is a limit in principle. Following the Weyl obstruction (Doc. 316), the spectrum of any compact resonance space grows as $N(\lambda) \sim \lambda^{d/2}$, while the arithmetic counting density grows as $T \ln T$. A finite resonance space cannot carry this density – independently of grid width and computational precision.

Primes as generators

[K] What actually carries the resonance structure are the primes. The Euler product treats each prime on its own, without cross terms; the dual length spectrum contains only $k \ln p$. The harmonic series shows the same structure physically: in the series $n f_0$, exactly the prime orders introduce new tonal qualities, while $4 = 2 \cdot 2$ and $6 = 2 \cdot 3$ are combinations.

A method that treats primes as orthogonal axes works structurally correctly. A method that treats a composite grid width as a resonance parameter does not.

.5 Balance

Statement	Status
Period finding is a spectral problem	[K]
Primes carry the resonance structure	[K] measured by bicoherence
Preparation accounts for ~95 % of gate cost	[K] (Doc. 176)
Sequential period search is $O(N)$	[K] measured
σ values are sampling grids	[K]
σ values generate eigenfrequencies	[X] composite
Classical period search reaches $O((\log N)^3)$	[X] not established
Twin-prime optimisation via grid width	[X] not reproducible
Scaling solvable by computational precision	[X] Weyl obstruction

The load-bearing sentence. The test series documents the behaviour of sequential period-search methods. It shows no advantage over established methods, and the observed differences between grid widths are sampling artefacts. The resonance description itself remains correct – but its carriers are the primes, not the grid.

.6 Relation to the Corpus

The placement draws on Doc. 316 (August 2026): bicoherence test, Weyl obstruction, and the role of primes as generators. The revision is declared under R75 in Doc. 190.

.7 Verification

Reproducible via `s1_periodensuche.py` (correctness, scaling, σ factorisation, comparison of methods) and `s2_grenzen.py` (limits of both approaches). Standard library only, target values as assertions, seed 20780458.